An SAM Fine-Tuning Framework With Frequency-Domain Interactive LoRA for Remote Sensing Change Detection

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***Abstract*—Achieving high-accuracy remote sensing change detection (RSCD) algorithms requires high-quality semantic feature extraction from remote sensing images (RSIs). Due to its powerful general-purpose feature extraction capability, the segment anything model (SAM) has found wide application across diverse fields. However, SAM may not be optimally suited for RSIs. To address this limitation, we propose a frequency-domain interactive low-rank adaptation (LoRA) fine-tuning architecture (FILFArch) to enhance the performance of SAM in RSCD tasks. Based on FILFArch, we then develop two task-specific algorithms: the FILFBCD for binary change detection (BCD) and the FILFSCD for semantic change detection (SCD). To enhance the capability of SAM in capturing bi-temporal RSIs feature relationship, the bi-temporal interaction fusion LoRA (BIF-LoRA) is designed with a Siamese architecture. Within BIF-LoRA, frequency-domain feature interaction (FDFI) utilizes fast Fourier transform block (FFTB) to fuse bi-temporal frequency-domain features. This enables cross-temporal frequency-domain interaction, effectively discriminating spatiotemporal feature differences. Additionally, we use a shared BCD Decoder to serve as the binary change detector for both FILFBCD and FILFSCD. The BCD Decoder first applies a coarse difference feature extraction (CDFE) to coarsely fuse deep semantic features, yielding a coarse-grained change feature map. Subsequently, a frequency-domain feature enhancement (FDFE) refines these abstract features to generate a fine-grained change map. In FILFSCD, FDFE is further utilized to recover the semantic change information of each temporal RSI. Experimental results demonstrate that FILFBCD achieves the highest $F1$ scores of 83.53%, 66.75%, and 83.79% on BCD datasets MLCD, S2Looking, and SYSU-CD, respectively. Meanwhile, FILFSCD achieves the highest $F1$ scores of 64.05% and 87.02% on SCD datasets SECOND and DSCD, respectively. These results demonstrate the effectiveness and versatility of the proposed FILFArch for RSCD tasks. The code is available at https://github.com/juncyan/filora**

***Index Terms*—Binary change detection (BCD), frequency-domain interactive LoRA, remote sensing change detection, segment anything model (SAM), semantic change detection (SCD).**

I. Introduction

REMOTE sensing change detection (RSCD) involves identifying changes on the Earth's surface using multitemporal satellite or aerial imagery [1]. RSCD plays an important role in environmental monitoring, urban expansion analysis, agricultural land use assessment, and so on [2], [3], [4]. With the rapid advancement of remote sensing technology, a large volume of high-resolution remote sensing images (RSIs) has been applied to RSCD [5]. However, effective comparison and merging of high-resolution RSIs to enable detailed and accurate detection of changes in land cover over temporal periods remains a significant challenge in RSCD tasks [6].

Depending on the granularity of the results, RSCD tasks are typically classified into binary change detection (BCD) and semantic change detection (SCD) [7]. BCD aims to classify land cover changes into binary labels by distinguishing changed pixels from unchanged ones in bi-temporal images [8]. In contrast, SCD not only identifies the locations of change but also assigns semantic categories, effectively modeling pixel-level "from–to" transitions [9]. Traditional RSCD methods accomplish inspection by manually designing feature change criteria, yet their generalization capability and robustness require enhancement. Deep learning-based RSCD methods automatically capture RSI information using deep neural networks, enabling effective handling of high-resolution images and complex tasks. Among deep learning-based methods, convolutional neural networks (CNNs) [10], Vision Transformer (ViT) [11], and Mamba [12] have been widely adopted for RSCD. CNNs have limited capability in modeling global contextual information. Due to the introduction of the self-attention mechanism, ViT can dynamically assign weights without being constrained by the receptive field, making them generally more effective in capturing global dependencies. However, self-attention [13] scales quadratically with input size, resulting in a substantial computational burden when processing high-resolution RSIs. VMamba [14] ingeniously integrates the strengths of the state space model (SSM) [15] and ViT, achieving global context modeling with linear computational complexity while preserving the ability to extract local

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features. However, VMamba integrates multiple components from both SSM and Transformer, resulting in a complex model architecture that poses significant challenges in training and hyperparameter tuning while exhibiting high sensitivity to task-specific variations.

Although the introduction of CNNs, Transformers, and Mamba models has advanced the development of RSCD algorithms, these models are typically trained on task-specific datasets, lacking broad visual knowledge and thus limiting their generalization ability in real-world applications [16]. Considering that RSCD tasks rely on the model's ability to extract generation information from RSIs, large-scale VFMs offer a new approach to addressing this issue [17]. Through pretraining on large-scale datasets, the segment anything model (SAM) [18] has developed powerful visual feature extraction capabilities, enabling it to generalize to unseen objects and scenes. However, due to the significant differences in shallow-level features of land cover objects caused by temporal spans, which lead to distinct visual characteristics in RSIs of different temporals, how to effectively transfer the capabilities of SAM to cross-temporal RSCD tasks has become a new research hotspot [6]. Today, fine-tuning techniques have focused primarily on adapting SAM for single-temporal image tasks such as segmentation and object detection [19], [20], [21]. These algorithms struggle to integrate and compare cross-temporal remote sensing features during feature extraction. In RSCD tasks, the primary approach leverages SAM's capability to extract and compare generic features from bi-temporal RSIs. Low-rank adaptation (LoRA) [22] introduces trainable rank-decomposition matrices to learn task-specific knowledge. This method exploits the synergy between task-specific and task-agnostic knowledge, enabling efficient fine-tuning. Therefore, LoRA is introduced to fine-tune the Image Encoder of SAM to learn RSCD task-specific knowledge [23] while ignoring the prompt and mask decoder, thereby achieving efficient fine-tuning.

The essence of RSCD is to accurately detect and quantify both subtle and large changes between bi-temporal RSIs. Existing spatial-domain algorithms (e.g., CNNs, Transformers, and VMamba) are highly sensitive to factors such as illumination variations and sensor noise between images. These nonland cover-related changes can be mistakenly identified as genuine changes, leading to misclassification [24]. In addition, they often struggle with irregular boundaries and edges in RSIs, failing to fully exploit critical change information. Moreover, they may overlook semantic inconsistencies across different feature levels, which hinders the accurate reconstruction of internal structures within changed objects [25]. To overcome the limitations of spatial-domain methods, researchers have begun to explore how to combine it with frequency-domain analysis to enhance the performance of RSCD [26]. Frequency-domain analysis transforms RSIs from the spatial domain to the frequency domain through a fast Fourier transform (FFT); this transformation reveals the spectral components of an image, which encode rich information about its structure and texture. By leveraging frequency-domain representations, models can enhance the extraction of fine details and edge features while effectively suppressing pseudo-changes caused by noise [27]. In the frequency domain, information such as the structure, texture, and patterns of images is represented by different frequency components [28]. Low-frequency components typically represent the overall contours and slowly varying regions of images, while high-frequency components contain details, edges, and rapidly changing regions. Deep learning methods primarily focus on contextual modeling of spatial-domain features in RSIs but have limitations in terms of frequency-domain knowledge, restricting further improvements in model performance [29]. Introducing frequency-domain information into RSCD can serve as an effective complement to spatial-domain features, providing a more comprehensive image description and thereby enhancing the model's ability to perceive subtle changes in bi-temporal RSIs.

In this article, we introduce a frequency-domain interactive LoRA to fine-tune SAM for RSCD tasks. Specifically, we propose a unified framework comprising FILFBCD for BCD tasks and FILFSCD for SCD tasks, collectively termed the frequency-domain interactive LoRA fine-tuning architecture (FILFArch). In FILFArch, bi-temporal interaction fusion LoRA (BIF-LoRA) is designed to fine-tune SAM as a feature extractor. BIF-LoRA employs a Siamese architecture with shared weights to process bi-temporal RSIs. Frequency-domain feature interaction (FDFI), serving as the core component of BIF-LoRA, leverages the FFT block (FFTB) to enable frequency-domain interaction between bi-temporal features. This facilitates the fusion of cross-temporal contextual feature correspondences from bi-temporal RSIs, thereby strengthening the consistency of bi-temporal feature representation. Additionally, the BCD Decoder is devised as a binary change detector for the RSCD tasks. BCD Decoder includes coarse difference feature extraction (CDFE) and frequency-domain feature enhancement (FDFE). CDFE is developed to perform frequency-domain guided bi-temporal fusion on the deep high-level features extracted by the SAM with BIF-LoRA, yielding coarse-grained change features. Subsequently, FDFE is employed to refine these features through frequency-domain guidance, producing fine-grained change maps with enhanced abstraction features. For SCD tasks, FDFE is further incorporated to recover the semantic change features. The contributions are as follows.

1) FILFArch is proposed to integrate cross-temporal feature relationships by leveraging FDFIs, aggregating the differential and similar features of bi-temporal RSIs, and enhancing the SAM's capability of extracting cross-temporal change features.
2) Based on the FILFArch architecture and incorporating the unique characteristics of RSCD tasks, we design the corresponding network for BCD and SCD tasks, respectively, named FILFBCD and FILFSCD.
3) BIF-LoRA adopts a Siamese weight-sharing architecture, where FDFI uses FFTB to extract consistent cross-temporal features in the frequency domain, ensuring efficient fine-tuning by leveraging these stable representations, thus enhancing the overall performance of FILFArch.

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4) A dedicated BCD Decoder is designed as a binary change detector. It employs the CDFE to integrate deep bi-temporal features and generate a coarse-grained binary change feature map. The FDFE then refines these representations under frequency-domain guidance to produce a fine-grained change map, enhancing the recovery of semantic details of change features. In particular, the FDFE is also adapted in FILFSCD as a semantic feature decoder.

II. Related Work

A. CNN-Based Methods

The objective of BCD algorithms is to achieve pixel-level identification of changed regions in RSIs. Following the emergence of FCNs [30] for image semantic segmentation (SS), Daudt et al. [31] pioneered the application of FCNs to BCD tasks. Subsequently, numerous representative network architectures have been introduced. Zhang et al. [32] proposed a Siamese structure for BCD, which is the dominant framework in the current RSCD tasks. In contrast, F3Net adopted a non-Siamese structure and used CNNs to filter channel-wise noise. However, because of its high computational cost, the non-Siamese structure remains nonmainstream. SNUNet [33] incorporates a more densely connected architecture that facilitates enhanced information exchange among shallow features derived from bi-temporal RSIs. FDCNN [34] leverages transfer learning to generate multiscale and multidepth difference feature maps. SAGNet [35] uses internal relationships among spatial context and cross-scale features in RSIs, effectively integrating both global and differential information for BCD. CD-RLKNet [36] applied reparameterized large-kernel convolutions to BCD, utilizing ultralarge CNNs with a kernel of 31×31 to achieve global feature fusion in RSIs.

BCD aims to localize changed regions between bi-temporal images and is well suited for scenarios requiring rapid identification of change areas [1]. However, it struggles to meet the demands of complex and fine-grained tasks such as urban planning and ecological assessment. SCD goes beyond simply identifying the changed regions by also classifying the semantic categories of land cover within those regions [9]. To address SCD tasks, Daudt et al. [9] proposed HRSCD, which employs an encoder with three independent branches: one for BCD and two for SS subtasks. Yang et al. [37] curated SECOND, a widely adopted benchmark dataset for SCD, and introduced a new evaluation metric to differentiate performance evaluation between BCD and SCD tasks. BiSRNet [38] proposed to mitigate insufficient information exchange between semantic and change decoders by introducing a novel loss function that strengthens the correlation between semantic branches. CdSC [39] employs 3-D convolution to capture spatiotemporal discrepancies in features and enhances consistency between differential features and bi-temporal representations through an integrated mechanism. Moreover, KPCA-MNet [40] explores unsupervised SCD approaches, extracting hierarchical features through pretrained CNNs or unsupervised learning techniques and subsequently comparing these features to identify changes. Wang et al. [41] proposed SiamContrast. SiamContrast adopts a Siamese architecture to effectively integrate SS techniques with FCNs, thereby enhancing the capability to detect changes in land cover. Guo et al. [42] proposed SC2A-SCD, a semi-supervised change detection method based on contrastive learning. By leveraging semi-supervised learning techniques, they effectively enhanced model performance in scenarios with limited RSIs.

Although CNN-based methods have significantly improved detection performance [43], their inherent limitation in modeling long-range dependencies among remote sensing features hinders their effectiveness in detecting large-scale or extensive changes [44]. To address these challenges, an increasing number of studies are exploring attention mechanisms to enhance feature representation and overall detection accuracy [45].

B. Transformer and Mamba-Based Methods

Transformer leverages attention mechanisms to model long-range dependencies among features, enabling effective capture of contextual relationships across distant image regions. ViT [11] marked a significant advancement by successfully adapting the Transformer to computer vision (CV) tasks. In the domain of RSCD, BIT [43] pioneered the integration of Transformer by tokenizing input images and modeling contextual spatial feature associations through self-attention mechanisms. ChangeFormer [46] adopted a Siamese ViT architecture specifically for BCD. ICIFNet [47] combined CNNs and Transformers to separately extract local and global features from bi-temporal RSIs, proposing an intracross interaction and interscale feature fusion network.

For Transformer-based SCD methods, SMNet [48] integrated CNNs and Transformers to extract semantic features from bi-temporal remote sensing imagery. It further enhanced performance through multitask branches and a custom-designed loss function. ChangeMask [49] further enhanced intertemporal feature interaction to model spatiotemporal relationships. SCanNet [50] employed a dual-branch encoder architecture that shares feature representations between the BCD and SS subtasks to explicitly learn semantic transitions.

Although Transformer-based methods effectively overcome the limited receptive field inherent in CNNs, they suffer from high computational costs. SSM [15] scales linearly with long-range sequence when modeling global contextual information. While Transformers exhibit quadratic computational complexity $O(N^2)$ with respect to sequence length N, SSM achieves linear complexity $O(N)$. Mamba [12], [51] has demonstrated strong performance as a backbone for language modeling. VMamba [14] extended the Mamba architecture to a wide range of CV tasks, demonstrating superior efficiency and accuracy.

RSMamba [52] introduced Mamba into RSCD and proposed a dynamic multipath activation mechanism to address Mamba's limitation of unidirectional modeling, achieving significant improvements in RSI classification. ChangeMamba [53] proposed three distinct feature scanning mechanisms as feature extractors to capture multistructural global contextual information from input RSIs. LCCDMamba [54] employed Siam-VMamba as its feature extractor and integrated strip

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convolutions to aggregate local change information in bi-temporal land cover features, thereby enhancing the detection of small objects and object boundaries. Zhou et al. [55] proposed MaDiNet, an SAR target detection method based on diffusion models. MaDiNet uses a Mamba-based diffusion model to improve both the quality of SAR data and the accuracy of target detection.

Current Mamba-based RSCD methods tailor different scanning strategies to serialize images according to the characteristics of RSIs, thus improving Mamba's ability to handle vision tasks [56]. However, most existing approaches still rely on row- or column-wise scanning, which introduces spatial discontinuities between adjacent pixels [57]. This limitation impedes the modeling of dependencies among neighboring pixels belonging to the same object [58], affecting the understanding of fine-grained changes.

C. SAM Application in Remote Sensing Tasks

Vision foundation models (VFMs) [59] are large-scale, general-purpose CV models trained on massive, diverse, and multisource datasets. These models exhibit strong zero-shot transfer capabilities for a wide range of downstream vision tasks, including image classification, object detection, and SS, without requiring task-specific retraining. SAM [18], a representative VFMs, has been widely adopted across domains such as autonomous driving, medical image analysis, and remote sensing [60], [61].

SAM constitutes a significant leap forward in capabilities for RSI interpretation and analysis. SAM-CD [62] leverages FastSAM [63] to extract bi-temporal change features and introduces a convolution adapter to aggregate these features for BCD, achieving efficient and accurate change identification. SCD-SAM [64] combines SAM with CNNs to jointly capture generic and local features from bi-temporal images. EFI-SAM [65] utilizes the pretrained Mobile SAM [66] to extract remote sensing features and designs a modulator to fuse spatial feature correlations from bi-temporal RSIs, producing coarse change maps; it then integrates multidimensional features to generate refined BCD results.

Existing methods primarily use the SAM Image Encoder as a fixed feature extractor and design auxiliary modules to improve the representation of remote sensing features, treating SAM primarily as a supporting backbone rather than an end-to-end trainable component [67]. Parameter-efficient fine-tuning (PEFT) is an effective strategy to adapt VFMs to specific downstream tasks without full retraining [22]. CWSAM [20] integrates lightweight adapters into SAM to enable efficient PEFT for SAR image segmentation. MeSAM [21] incorporates an Inception mixer into the SAM's Image Encoder to preserve high-frequency details, improving performance in optical image segmentation. SAM MLoRA [68] uses multiple parallel LoRA to enhance the low-rank matrix decomposition capabilities for object extraction. However, these PEFT-based approaches are primarily designed for single-temporal remote sensing tasks such as segmentation and object detection [69].

D. FFT Application in Remote Sensing Tasks

RSCD aims to identify changes in the position, extent, and nature of surface features through multitemporal RSIs [3]. In recent years, the combination of deep learning algorithms and FFT has made significant breakthroughs in RSCD tasks [24]. By mapping images from the spatial domain to the frequency domain, FFT can reveal hidden periodic features and noise distribution, while deep learning endows models with powerful feature learning capabilities. The synergy between these two approaches significantly improves the accuracy and robustness of change detection, particularly excelling in complex scenarios [29].

Chen et al. [70] proposed ProFiT for hyperspectral video tracking in RSIs, which significantly improves tracking accuracy by integrating frequency-aware filtering. FASI-Net [28] leverages frequency-domain features to suppress change-irrelevant components and enhance potential change regions, followed by fusing frequency- and spatial-domain features to generate the final change detection map. DDLNet [25] extracts frequency-domain features in addition to traditional spatial-domain features, using FFT to obtain RSI spectral information, and effectively combines both to capture comprehensive change information. Yang et al. [71] proposed EHCTNet integrates CNNs and Transformer for RSI feature extraction, introducing FFT to refine frequency-domain features. Chen et al. [72] proposed BPFT innovatively combining FFT with Transformer to simultaneously learn spatial–spectral fused features and global fused features, fully leveraging the complementarity of multimodal data. Jiang et al. [73] proposed that SFHFormer incorporates FFT into the Transformer. SFHFormer's spatial-domain branch captures local features, while its frequency-domain branch extracts global frequency features. FRCA [74] improves model precision by prioritizing high-frequency details by extracting frequency-domain information via FFT.

In this article, we combine FFT and LoRA fine-tuning algorithms, with the aim of enhancing detection accuracy and robustness through frequency-domain feature refinement and global–local modeling. We propose FILFArch, a novel LoRA framework specifically designed for RSCD tasks, which enables effective and efficient adaptation of SAM to the unique challenges of temporal feature alignment, change modeling, and semantic transition learning.

III. Methods

A. Overview

The framework proposed in this article focuses on two subtasks in RSCD tasks: BCD and SCD. BCD is the most extensively studied task. Specifically, it focuses on determining whether changes have occurred in the objects covered by RSIs. For the BCD task, given an input image pair (Img$A$, Img$B$), a binary change map (bc map) is generated. bc map $\in \{0, 1\}^{H\times W}$ represents a change map that accurately reflects the binary information of "change/no-change." SCD not only pays attention to the "location" where changes occur but also focuses on the semantic characteristics of the changes, namely, the "from–to" semantic change information. For SCD tasks, given

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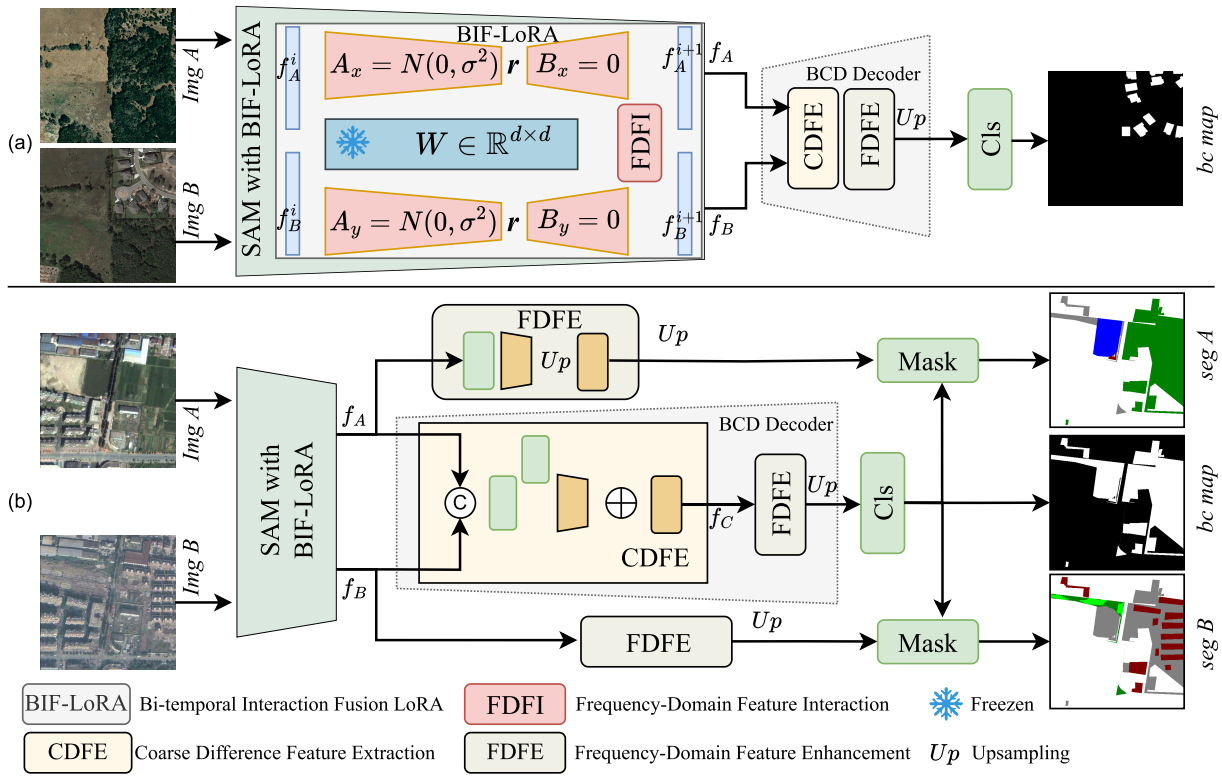


Fig. 1. Structure of the proposed FILFArch. (a) FILFBCD for BCD and (b) FILFSCD for SCD. In FILFArch, to enhance the capability of SAM in extracting cross-temporal remote sensing features while reducing training weights, BIF-LoRA is designed to extend LoRA to bi-temporal feature processing. BIF-LoRA employs frequency-domain guidance to fuse cross-temporal correlated features, thereby strengthening the extraction of change-related features. The BCD Decoder consists of the CDFE and FDFE. The CDFE performs an initial fusion of bi-temporal feature differences to generate coarse-grained change features. The FDFE leverages frequency-domain guidance to restore refined feature representations. Meanwhile, the FDFE is also utilized to recover semantic features in FILFSCD.

an input image pair (Img$A$ and Img$B$), it generates seg$A$, seg$B$, and bc map. seg$A$, seg$B$ $\in \{0,1,\ldots,C\}^{H\times W}$, where C represents the number of categories, are semantic information maps that predict the land cover of the bi-temporal RSIs. By combining seg$A$, seg$B$, and bc map, the "from–to" semantic change information can be obtained.

To address both BCD and SCD tasks, we propose FILFArch, termed FILFBCD and FILFSCD for the respective tasks. Fig. 1 illustrates FILFBCD and FILFSCD. FILFArch employs the SAM to fully extract robust and representative features from the input RSIs, which benefit from the SAM's general feature extraction capability. BIF-LoRA is designed to fine-tune SAM as a feature extractor. BIF-LoRA employs a Siamese structure with shared weights to process bi-temporal remote sensing information, enhancing the model's capability to extract cross-temporal feature correlations. FDFI, as the core component of BIF-LoRA, utilizes FFTB to fuse frequency-domain features to compare bi-temporal feature differences, extract more consistent change-related information, and achieve cross-temporal feature alignment and fusion. The SAM with BIF-LoRA processes bi-temporal RSIs simultaneously, enhancing the model's ability to capture cross-temporal feature correlations and generating the feature maps f_A and f_B, respectively.

Meanwhile, the BCD Decoder serves as a binary change detector for both FILFBCD and FILFSCD and is designed to capture spatiotemporal change relationships within the features extracted by SAM with BIF-LoRA. The BCD Decoder is divided into CDFE, which integrates change information, and FDFE, which is responsible for recovering abstract semantic information. Furthermore, FDFE acts as the semantic decoder for the detection of the land cover category in FILFSCD, generating semantic change features. Finally, FILFSCD outputs two semantic maps and a binary change map. In FILFBCD, f_A and f_B are processed through the BCD Decoder and Cls to obtain bc map, where Cls is a convolution layer. Meanwhile, in FILFSCD, f_A is passed through the semantic decoder and a mask to obtain s_A to produce the segmentation result seg A. The mask is as follows:

$$s_A = \text{FDFE}(f_A) \tag{1}$$

$$\text{seg}A = \text{Conv}(s_A) \times \text{bc map} \tag{2}$$

where Conv indicates convolution. Similarly, seg$B$ is calculated using the same method applied to f_B.

B. Bi-Temporal Interaction Fusion LoRA

SAM adopts a ViT-based architecture, which comprises multiple ViT blocks to progressively extract and abstract the features of RSIs. The structure of the ViT block is illustrated in Fig. 2(a). BIF-LoRA is applied to both the attention block and the MLP within the ViT block. LoRA is a PEFT approach, and the standard LoRA is illustrated in Fig. 2(b). For a linear

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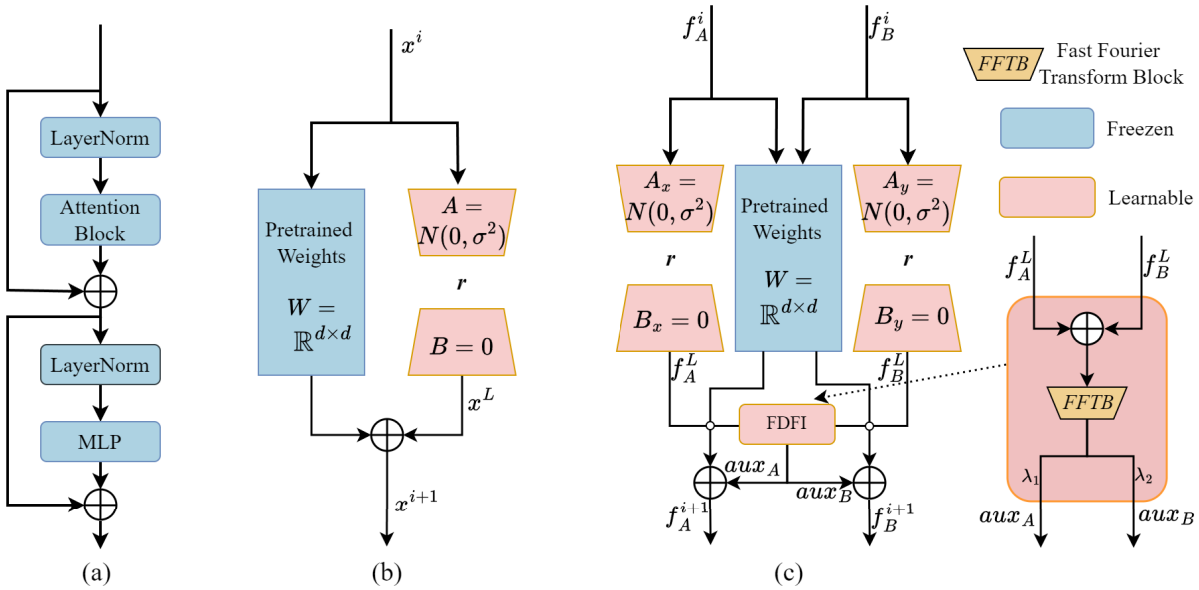


Fig. 2. Details of SAM adaptation methods. (a) Standard ViT block, (b) standard LoRA, and (c) proposed BIF-LoRA. Adapters facilitate the efficient utilization of general knowledge in RSCD tasks. Compared to standard LoRA, BIF-LoRA introduces a Siamese architecture to process bi-temporal features. Moreover, frequency-domain fusion is employed to integrate these features, thereby enhancing SAM's ability to discriminate fine-grained details.

layer with an input x^i and weight matrix $W \in \mathbb{R}^{d\times d}$, LoRA assumes that the update weights can be approximated by the product of two low-rank matrices

$$\Delta W = BA \tag{3}$$

where $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times d}$, and $r \ll d$. After applying LoRA, the output of the linear layer can be expressed as follows:

$$x^{i+1} = (W + \Delta W) \times x^i = W \times x^i + B\left(A \times x^i\right) \tag{4}$$

where A is initialized with a random Gaussian distribution and B is initialized to zero. During the training process, the original weight W remains frozen. Only the low-rank matrices A and B are involved in the training.

BIF-LoRA extends the standard LoRA to handle RSCD tasks by adapting to bi-temporal input features. The architecture of BIF-LoRA is illustrated in Fig. 2(c). BIF-LoRA employs a Siamese design that shares the frozen SAM backbone while introducing branch-specific LoRA adapters (with rank $r = 8$) for each temporal RSIs. To further strengthen temporal coherence, FDFI enhances cross-temporal feature correlations through frequency-domain modeling, generating bi-temporal auxiliary features aux. FDFI utilizes the FFT algorithm, and frequency-domain operations achieve high computational efficiency, making them particularly suitable for processing complex remote sensing backgrounds. To emphasize the independence of features from different temporal modalities, λ_1 and λ_2 are employed as temporal auxiliary coefficients. Here, λ_1 and λ_2 are learnable parameters used to modulate the weights of individual temporal features in the fused frequency-domain feature representation. Both are initialized to 0.5, indicating equal temporal weighting at the initial stage. The aux computational is as follows:

$$f_A^L = B_x\left(A_x \times f_A^i\right) \tag{5}$$

$$f_B^L = B_y\left(A_y \times f_B^i\right) \tag{6}$$

$$f_F = \mathrm{FFTB}\left(f_A^L + f_B^L\right) \tag{7}$$

$$\mathrm{aux}_{A/B} = \lambda_{1/2} \times f_F \tag{8}$$

where FFTB indicates the FFTB. Finally, the output of BIF-LoRA is obtained by fusing the cross-temporal frequency-domain interactive features aux, with the computation formulated as follows:

$$f_{A/B}^{i+1} = W \times f_{A/B}^i + \mathrm{aux}_{A/B}. \tag{9}$$

C. BCD Decoder

The BCD Decoder is a shared component in both FILFBCD and FILFSCD. It takes as input the encoded features f_A and f_B from the bi-temporal RSIs and outputs the final binary change map bc map. The structure of BCD Decoder is illustrated in Fig. 3, which comprises CDFE and FDFE. CDFE lies in initially integrating the feature differences between bi-temporal RSIs to generate a relatively coarse yet comprehensive feature representation. This representation is able to capture major change trends and large-scale structural variations. Building upon this foundation, FDFE refines these coarse-grained features, enhancing their detailed expressiveness. This results in a more precise and detailed change map while also recovering more semantic information. Through this collaborative approach, where CDFE and FDFE work in tandem, a coarse-to-fine change detection process is achieved, ensuring the accuracy and completeness of the detected change features. Furthermore, in FILFSCD, FDFE also serves as the semantic decoder for land cover classification, producing semantic information features to support SCD tasks.

1) Coarse Difference Feature Extraction: In CDFE, the encoded features f_A and f_B are first concatenated and fed into a convolution to generate preliminary fused features

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f_D. Subsequently, the FFTB is employed to capture periodic structural patterns and long-range correlation characteristics of f_D. Specifically, FFT is utilized to map spatial features into the frequency domain, where learnable weights L_W are applied to differentially amplify frequency components, enabling self-attention across frequency bands. The enhanced frequency-domain features are then transformed back into the spatial domain using the inverse FFT (IFFT). The computational process is as follows:

$$f_D = \text{Conv}\left(\text{cat}\left(f_A, f_B\right)\right) \tag{10}$$

$$f_S = \mathcal{F}^{-1}\left(L_W \times \mathcal{F}\left(f_D\right)\right) \tag{11}$$

where Conv indicates convolution, $\mathcal{F}$ indicates fast Fourier transformation, and $\mathcal{F}^{-1}$ indicates inverse fast Fourier transformation.

In the FFTB, the FFT decomposes image features into spectral components with distinct physical interpretations: low-frequency components represent global semantic structures and illumination, while high-frequency components encode fine-grained edges and textures. Meanwhile, learnable weights L_W are applied to selectively amplify or suppress these frequency components. This mechanism is effectively functioning as an adaptive spectral attention module. It allows the FFTB to focus on structure, preserving low frequencies, highlighting boundaries and high frequencies while attenuating noise. This enables robust self-attention across frequency bands. Meanwhile, f_D undergoes processing through a channel attention mechanism to capture global change features in the spatial domain that generate f_S'. The computational process is as follows:

$$f_S' = f_D \times \text{BN}\left(\text{Conv}\left(\text{GAP}\left(f_D\right)\right)\right) \tag{12}$$

where BN indicates batch normalization and GAP indicates global average pooling.

Finally, the refined feature map f_D' is processed through a feed-forward network (FFN) [11] to generate coarse-grained difference features f_C. FFN further refines the features by recovering fine-grained details, thereby enhancing spatial consistency. It is worth noting that the FFN is a preexisting component within the ViT and is not specifically designed in this work.

2) Frequency-Domain Feature Enhancement: FDFE aims to improve coarse-grained change features f_C and reconstruct them into a binary change map bc map. Its computational process is as follows:

$$\text{bcd map} = \text{FFN}\left(\text{Up}\left(\text{FFTB}\left(\text{Conv}\left(f_C\right)\right)\right)\right). \tag{13}$$

FDFE performs adaptive multiscale feature fusion by applying learnable weights to different frequency components. In the BCD Decoder, the FDFE is utilized to enhance the coarse-grained change features generated by the CDFE and reconstruct them into fine-grained change maps. In the SCD task, the FDFE also serves as a semantic feature restoration module, improving the recovery capability of semantic details through frequency-domain enhancement techniques, thus improving the accuracy of land cover classification.

D. Loss Function

This article primarily investigates the performance of the proposed architecture on the RSCD tasks. Consequently, instead of designing task-specific loss functions for BCD and SCD, we optimized the network using commonly adopted loss functions within the field.

For the BCD task, the binary cross-entropy loss function is selected, with its computational formula defined as follows:

$$L_{\text{BCD}} = -y_C \log\left(p_C\right) - \left(1 - y_C\right) \log\left(1 - p_C\right) \tag{14}$$

where y_C represents the ground-truth change label and p_C represents the predicted probability of the change targets.

For SCD tasks, we select the hybrid loss function originally proposed in BiSRNet [38]. This hybrid loss function is the predominant choice for training SCD models. This loss function contains three parts: BCD loss L_{BCD}, SS loss L_{ss}, and semantic consistency loss L_{sc}. The calculation is as follows:

$$L_{\text{ss}} = -\sum_{i=1}^{N} y_i \log\left(p_i\right) \tag{15}$$

where N denotes the number of categories, y_i represents the probability of the target class, and p_i indicates the probability of the predicted class

$$L_{\text{sc}} = \begin{cases} 1 - \cos\left(x_A, x_B\right), & y_C = 1 \\ \cos\left(x_A, x_B\right), & y_C = 0 \end{cases} \tag{16}$$

where x_A and x_B are feature vectors of a pixel on seg$A$ and seg$A$, respectively. Here, cosine similarity is used to compute the consistency loss. By measuring the cosine of the angle between two feature vectors, cosine similarity effectively evaluates their directional consistency rather than merely their numerical closeness. This property enables the model to better capture the consistency of the same object's features at different temporal points, thereby enhancing its sensitivity to subtle changes.

The overall loss function is obtained by summing up the three individual loss functions

$$L_{\text{SCD}} = 0.5 \times \left(L_{\text{ss}A} + L_{\text{ss}B}\right) + L_{\text{BCD}} + L_{\text{sc}} \tag{17}$$

where $L_{\text{ss}A}$ and $L_{\text{ss}B}$ are L_{ss} of each temporal branch.

IV. Experiments

A. Datasets

These studies evaluate on three datasets for BCD tasks: MLCD [65], S2Looking [75], and SYSU-CD [76], as well as two SCD datasets: SECOND [37] and DSCD.

1) *MLCD* [65] comprises 10 000 pairs of RSIs collected from the Google Earth Engine, spanning the period from 2008 to 2023. It includes RSIs documenting land–sea changes and vegetation dynamics in the Macao SAR.
2) *S2Looking* [75] includes 5000 pairs of RSIs from rural areas around the world. S2Looking includes over 65 920 annotated change instances, which mark new constructions and demolitions within each sample.

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TABLE I
COMPARISON WITH SOTA NETWORKS FOR BCD. THE BEST METRICS ARE MARKED IN BOLD, AND THE SECOND BEST IS UNDERLINED

Models	MLCD				S2Looking				SYSU-CD			
	OA(%)	IoU(%)	Kappa(%)	F1(%)	OA(%)	IoU(%)	Kappa(%)	F1(%)	OA(%)	IoU(%)	Kappa(%)	F1(%)
USSFCNet [78]	87.73	56.34	64.33	72.07	90.94	38.42	50.87	55.51	90.67	66.94	74.09	80.19
CD-RLKNet [36]	88.57	58.91	66.91	74.14	99.19	43.84	60.55	60.95	90.89	68.82	72.46	81.39
RS-Mamba [52]	89.93	63.70	71.36	77.83	99.16	43.72	60.42	60.84	89.71	62.30	70.20	76.77
DGMAANet [79]	90.13	64.57	72.10	78.47	91.36	40.28	52.99	57.42	91.47	69.17	76.21	81.77
SAM-CD [62]	90.26	65.16	72.60	78.91	99.06	39.60	56.27	56.71	90.50	66.81	73.87	80.10
LCCDMamba [54]	88.88	61.01	68.61	75.78	99.24	45.71	62.08	62.72	91.40	67.59	75.15	80.66
Clafr-L0 [80]	89.12	62.74	69.98	77.10	91.51	39.90	52.65	57.04	90.03	64.76	72.11	78.61
ISDANet [81]	86.60	58.59	64.90	73.89	99.05	43.17	59.95	60.31	90.94	66.08	73.06	80.62
EFI-SAM [65]	90.87	67.72	74.77	80.75	99.20	47.85	64.41	64.74	92.18	70.74	77.16	82.86
FILFBCD (ViT-H)	91.01	68.32	75.39	81.18	99.23	48.47	64.90	65.29	91.66	69.16	76.33	81.77
FILFBCD (Tiny-ViT)	91.78	**71.74**	78.06	83.34	**99.26**	**50.09**	**66.38**	**66.75**	92.13	71.69	78.34	83.51
FILFBCD (Hiera-Large)	**92.12**	71.73	**78.36**	**83.53**	**99.26**	49.63	65.97	66.34	**92.56**	**72.10**	**78.96**	**83.79**

3) *SYSU-CD* [76] collects Hong Kong RSIs between 2007 and 2014, showcasing the evolution of high-rise buildings and ports. However, its wide range of change types introduces more distracting information, making change detection more challenging.
4) *SECOND* [37] provides a collection of 4662 pairs of 512×512 aerial images, with spatial resolutions ranging from 0.5 to 3 m per pixel. The dataset includes six classes: nonvegetated ground (N.v.g) surfaces, trees, low vegetation, water, buildings, and playgrounds, aiming to capture a diverse range of geographical changes.
5) *DSCD* is our annotated land cover SCD dataset. This dataset encompasses desert changes in and around the Mu Us Desert from 2008 to 2023, with a total of four categories labeled: water, woodland, farmland, and desert.[1]

B. Implementation Details and Evaluation Metrics

To validate the performance of FILFArch, we adopt three different Image Encoders, from SAM-H (ViT-H) [18], Mobile SAM (Tiny-ViT) [66], and SAM2.1-Hiera-Large (Hiera-Large) [77], as encoder. Consequently, both FILFBCD and FILFSCD are provided in three versions: ViT-H, Tiny-ViT, and Hiera-Large.

All experiments are trained on RTX4090D. The AdamW optimizer is used for training, with an initial learning rate set to $3e^{-5}$. The maximum number of epochs is set to 200. A Cosine annealing decay is used for adjusting the learning rate.

To evaluate the performance of the model, task-specific metrics are selected according to the requirements of different RSCD tasks. For BCD tasks, overall accuracy (OA), intersection over union (IoU), $F1$ score ($F1$), and Kappa coefficient (Kappa) are used, with computation detailed in [36]. For SCD tasks, OA, mean IoU (MIoU), separated Kappa (Sek), and $F1$ are selected, with calculation formulas referenced from [38].

C. BCD Studies

To demonstrate the performance advantages of FILFBCD, USSFCNet [78], CD-RLKNet [36], RS-Mamba [52], DGMAANet [79], SAM-CD [62], LCCDMamba [54], Clafr-L0 [80], ISDANet [81], and EFI-SAM [65] are selected for comparison. Among these compared methods, SAM-CD employs Fast SAM [63] as its backbone, and EFI-SAM adopts Mobile SAM as its backbone.

Table I presents the quantitative results of BCD studies, which clearly demonstrate that FILFBCD significantly outperforms comparison methods. Specifically, all three variants of FILFBCD rank among the top three on both MLCD and S2Looking. On MLCD, the IoUs are 68.32%, 71.74%, and 71.73%. On S2Looking, the IoU are 48.47%, 50.09%, and 49.63%, respectively. On the SYSU-CD dataset, FILFBCD (Hiera-Large) and FILFBCD (Tiny-ViT) achieve the top two rankings, with IoU scores of 71.69% and 72.10%, respectively. Only FILFBCD (ViT-H), with IoU of 69.16%, is slightly lower than that of EFI-SAM with IoU of 70.74%. The OA, Kappa, and $F1$ exhibit consistent trends with IoU.

Fig. 4 presents the prediction results on MLCD, S2Looking, and SYSU-CD. The visualization results clearly demonstrate that all three variants of FILFBCD effectively detect changes of diverse types, scales, and sizes across various scenarios. In particular, in complex scenes, such as the first row of MLCD and the second row of SYSU-CD, FILFBCD (Tiny-ViT) and FILFBCD (Hiera-Large) accurately capture intraclass differences and fine edge details. This success is attributed to the FDFI capability of BIF-LoRA, which improves SAM's ability to extract global contextual information from cross-temporal features. Consequently, FILFBCD successfully identifies consistent features across temporal RSIs. These results comprehensively validate FILFBCD's robust feature extraction and representation capabilities under diverse and challenging conditions, further confirming that our FDFI fine-tuning framework can effectively leverage SAM's general-purpose knowledge to facilitate BCD.

D. SCD Studies

To demonstrate the performance advantages of FILFSCD, HRSCD4 [9], SSCDl [38], BiSRNet [38], SSESN [82], CdSC [39], MambaSCD [53], SCD-SAM [64], and SAM-FE [83] are selected for comparison. Among these compared methods, both SCD-SAM and SAM-FE use Mobile SAM as their backbone.

[1]It can be accessed at https://www.modelscope.cn/datasets/chuntsing/DSCD

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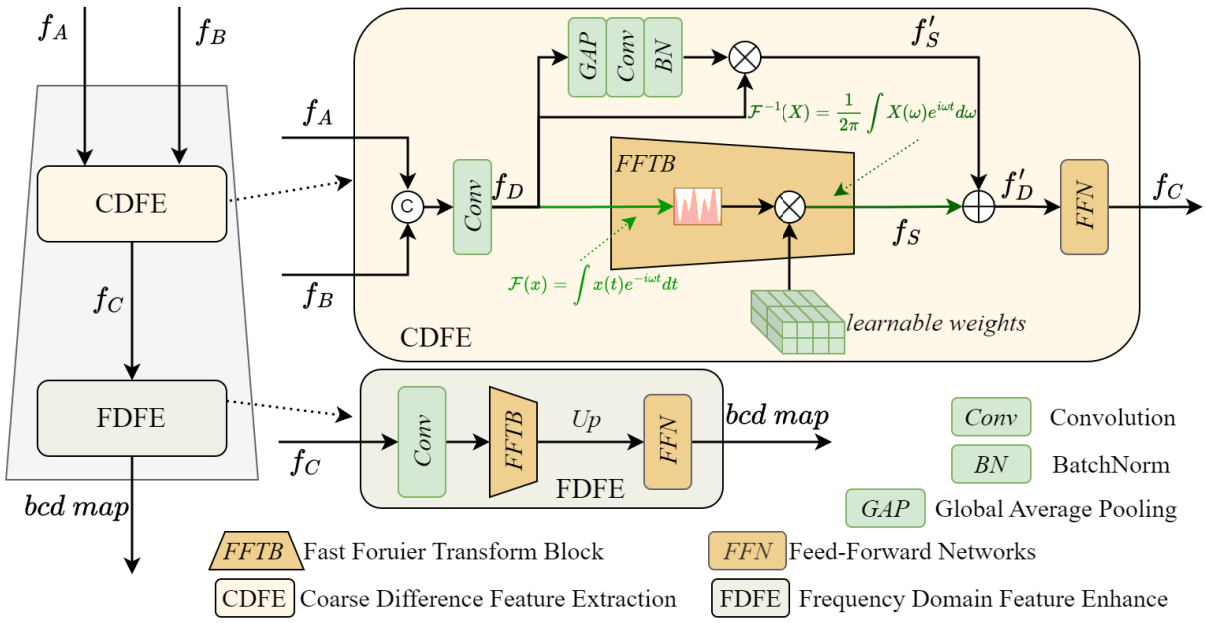


Fig. 3. Structure of the proposed BCD Decoder, which consists of the CDFE and the FDFE. The CDFE compares bi-temporal features by first mapping them into the frequency domain via the FFT. It analyzes multiscale feature differences in the frequency domain to generate coarse-grained change features. The FDFE utilizes the FFTB to convert low-resolution change features from spatial to frequency domain. By capturing multiscale and long-range contextual information, it enables more accurate feature recovery through global dependency modeling.

TABLE II

COMPARISON WITH SOTA NETWORKS FOR SCD. THE BEST METRICS ARE MARKED IN BOLD, AND THE SECOND BEST IS UNDERLINED

Methods	SECOND				DSCD			
	OA(%)	MIoU(%)	Sek(%)	F1(%)	OA(%)	MIoU(%)	Sek(%)	F1(%)
HRSCD4 [9]	88.65	70.71	18.07	58.31	86.30	74.71	35.92	72.99
SSCDl [38]	86.13	71.22	19.82	59.94	86.86	75.64	40.52	77.30
BiSRNet [38]	86.77	71.60	20.30	60.71	88.23	78.18	46.25	80.79
SSESN [82]	89.04	72.00	21.41	61.97	88.72	79.15	47.03	82.67
CdSC [39]	89.18	72.25	20.20	60.62	88.60	78.64	45.21	79.36
MambaSCD [53]	87.98	<u>73.25</u>	23.15	63.08	90.79	82.64	54.69	84.60
SCD-SAM [64]	86.87	67.18	13.22	53.77	83.44	70.47	30.97	70.17
SAM-FE [83]	<u>89.82</u>	73.08	22.67	63.41	89.33	80.01	48.14	83.16
FILFSCD (ViT-H)	89.57	72.89	22.63	63.12	90.98	82.89	56.11	85.87
FILFSCD (Tiny-ViT)	**89.85**	**73.39**	**23.73**	**64.05**	<u>91.55</u>	<u>83.86</u>	<u>58.09</u>	<u>86.75</u>
FILFSCD (Hiera-Large)	89.81	73.21	<u>23.33</u>	<u>63.65</u>	**91.66**	**84.05**	**58.57**	**87.02**

Table II presents the quantitative results of the SCD studies, which demonstrate the superior performance of FILFSCD, with four metrics. FILFSCD (Tiny-ViT) achieves the highest metrics on SECOND, with OA of 89.58%, MIoU of 73.39%, Sek of 23.73%, and $F1$ of 64.05%. FILFSCD (Hiera-Large) ranks second on SECOND in Sek of 23.33% and $F1$ of 63.65%. On DSCD, FILFSCD (Hiera-Large) performs the best, achieving OA of 91.66%, MIoU of 84.05%, Sek of 58.57%, and $F1$ of 87.02%. The other two variants, FILFSCD (Tiny-ViT) and FILFSCD (ViT-H), rank second and third, respectively, with all metrics slightly lower than those of FILFSCD (Hiera-Large).

Figs. 5 and 6 illustrate the visualization results in the SECOND and DSCD, respectively. As evidenced by the visualization results, in complex urban land cover scenarios, such as the second row of Fig. 5, FILFSCD (Tiny-ViT) and FILFSCD (Hiera-Large) produce notably sharper and more accurate boundary predictions. In Fig. 6, when changes involve large-scale transitions such as desertification or water body shrinkage, FILFSCD consistently assigns correct semantic labels across temporally variant regions. Specifically, in the first row of Fig. 6, FILFSCD series accurately identifies the transition from farmland to desert while preserving spatial coherence. This highlights the effectiveness of the proposed BIF-LoRA, which effectively models land cover semantics by leveraging frequency-domain interactions, even in challenging conditions like low contrast and complex environments. By integrating frequency-domain cross-temporal interactions, FILFSCD successfully adapts the SAM to the SCD tasks, thereby achieving more accurate and reliable SCD in bi-temporal RSIs.

E. Efficient Comparison

To comprehensively evaluate and compare the computational efficiency of various models, we adopt four key metrics: number of parameters (Params., in millions), floating-point

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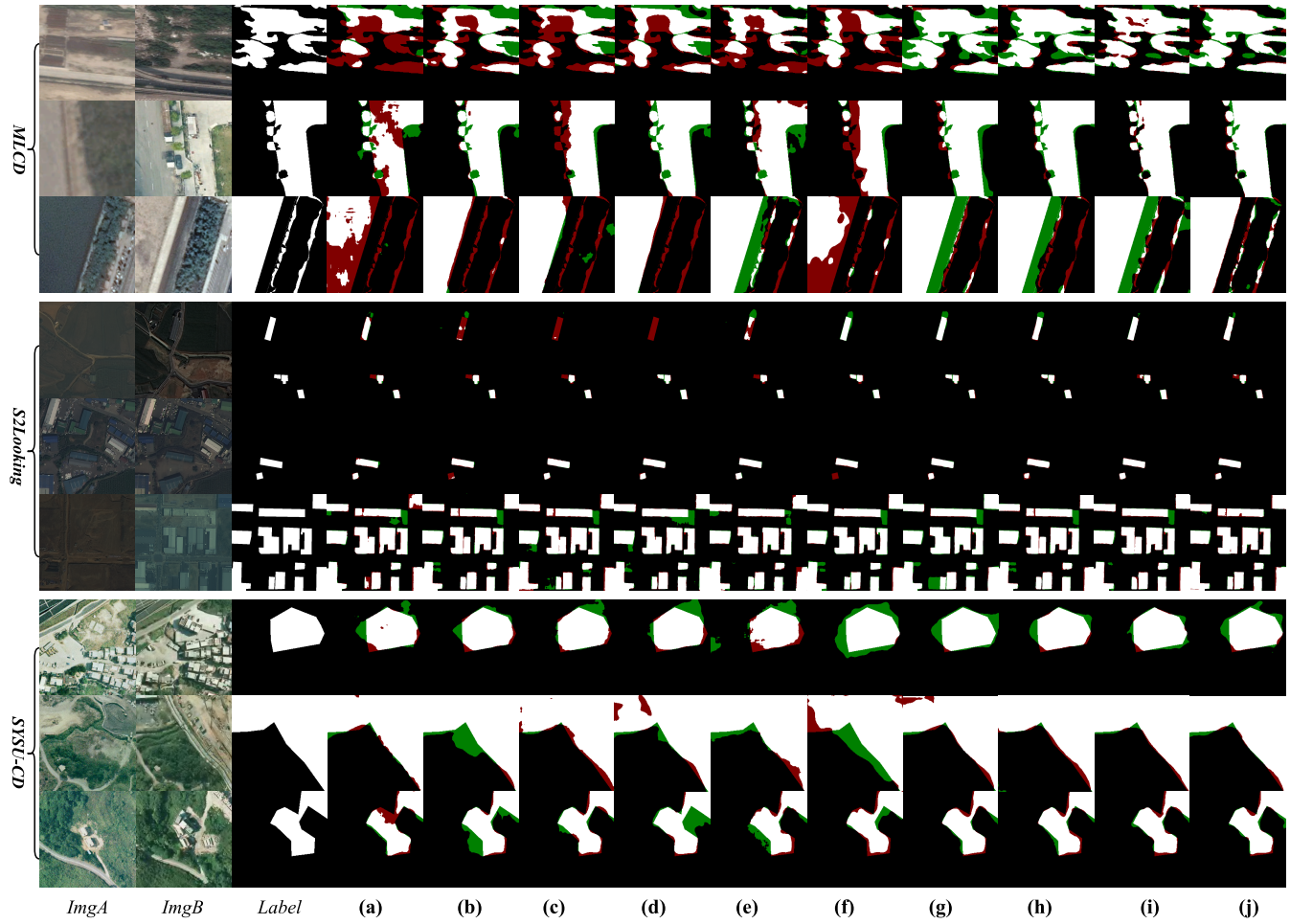


Fig. 4. Visualization of BCD, where white indicates TP, black indicates TN, green indicates FP, and red indicates FN. (a)–(g) RS-Mamba [52], DGMAANet [79], SAM-CD [62], LCCDMamba [54], Clafr-L0 [80], ISDANet [81], and EFI-SAM [65], respectively. (h)–(j) Our proposed FILFBCD (ViT-H), FILFBCD (Tiny-ViT), and FILFBCD (Hiera-Large), respectively.

TABLE III
COMPARISON OF BCD MODELS IN COMPUTATIONAL COST

BCD Methods	Params.(M)	Flops(G)	VRAM(G)	Infer.(ms)
USSFCNet [78]	1.52	4.86	0.19	0.41
CD-RLKNet [36]	31.25	47.24	0.20	2.15
RS-Mamba [52]	42.30	18.36	0.38	2.04
DGMAANet [79]	37.10	18.10	0.29	0.47
SAM-CD [62]	2.59	8.60	0.48	2.39
LCCDMamba [54]	93.90	38.20	1.13	2.07
Clafr-L0 [80]	4.18	16.41	0.20	1.76
ISDANet [81]	6.40	3.80	0.45	1.51
EFI-SAM [65]	5.83	6.17	0.10	0.94
FILFBCD (ViT-H)	631.58*+7.02	183.89	4.77	9.31
FILFBCD (Tiny-ViT)	5.75†+0.92	3.64	0.06	0.59
FILFBCD (Hiera-Large)	213.49‡+4.23	101.39	7.58	6.81

* SAM-H
† Mobile SAM
‡ SAM2.1-Hiera-Large

TABLE IV
COMPARISON OF SCD MODELS IN COMPUTATIONAL COST

Methods	Param.(M)	Fops(G)	VRAM(G)	Infer.(ms)
HRSCD4 [9]	13.71	10.98	0.21	2.96
SSCDl [38]	23.31	47.44	0.23	2.50
BiSRNet [38]	23.38	47.54	0.24	1.48
SSESN [82]	51.31	69.67	0.59	1.78
CdSC [39]	33.86	33.70	0.55	3.40
MambaSCD [53]	90.12	53.93	2.19	9.45
SCD-SAM [64]	117.02	62.64	4.67	6.48
SAM-FE [83]	17.37	20.16	0.43	1.55
FILFSCD (ViT-H)	631.58*+7.20	184.36	4.77	9.33
FILFSCD (Tiny-ViT)	5.75†+1.09	4.76	0.07	0.68
FILFSCD (Hiera-Large)	212.69‡+4.49	101.86	7.60	6.85

* SAM-H
† Mobile SAM
‡ SAM2.1-Hiera-Large

operations (Flops, in gigaflops), peak video RAM (VRAM, in gigabytes), and per-image inference time (Infer., in milliseconds). Tables III and IV present the efficiency metrics for BCD models and SCD models, respectively. All efficiency metrics were measured with a consistent input resolution of 256×256 pixels.

As shown in Table III, FILFBCD (ViT-H) with Params. of 638.6 M, which is 7.02 M more than SAM-H. Similarly, FILFBCD (Tiny-ViT) contains 0.92 M more than Mobile SAM, and FILFSCD (Hiera-Large) exceeds SAM2.1-Hiera-Large by 4.23 M. A consistent trend is also observed in Table IV, indicating that the model size of the proposed FILFArch is closely aligned with that of its underlying VFMs.

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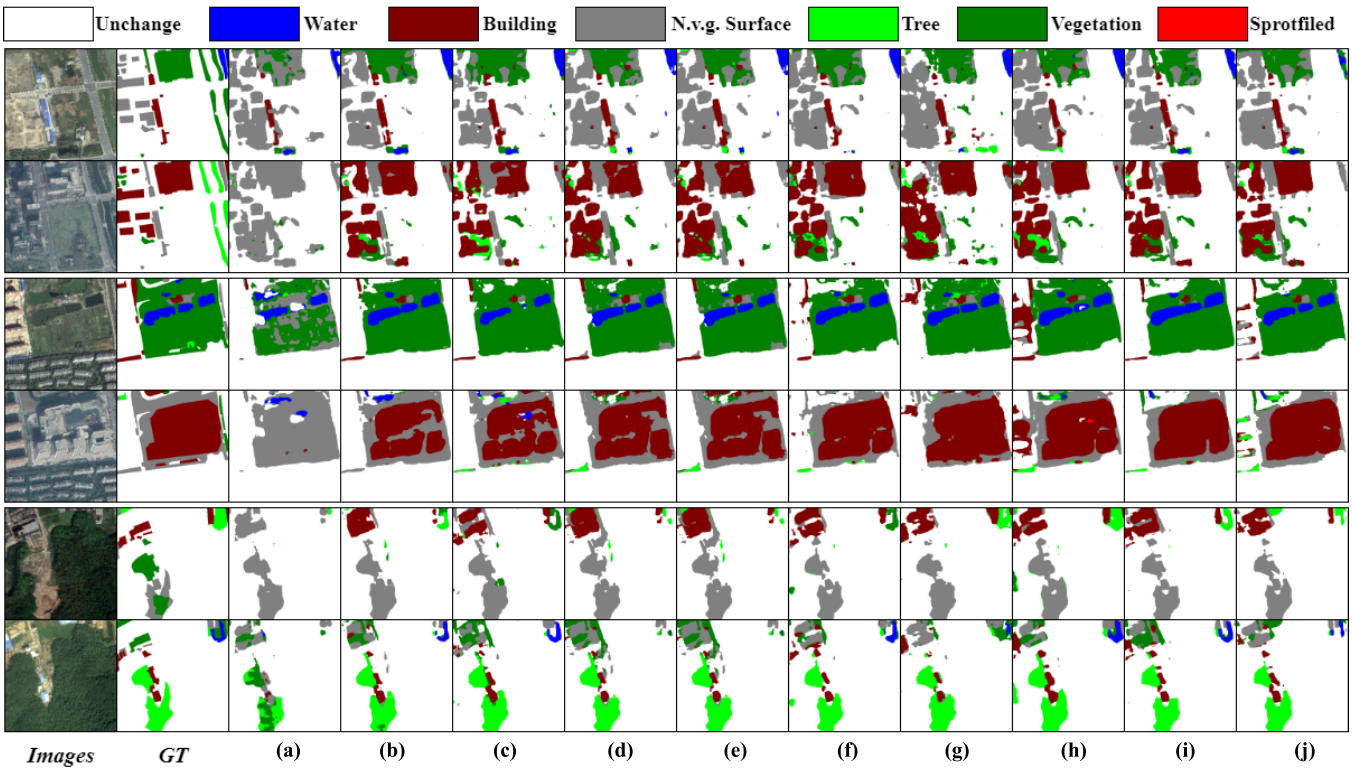


Fig. 5. Visualization of SECOND. (a)–(g) SSCDI [38], BiSRNet [38], SSESN [82], CdSC [39], MambaSCD [53], SCD-SAM [64], and SAM-FE [83], respectively. (h)–(j) Our proposed FILFSCD (ViT-H), FILFSCD (Tiny-ViT), and FILFSCD (Hiera-Large), respectively.

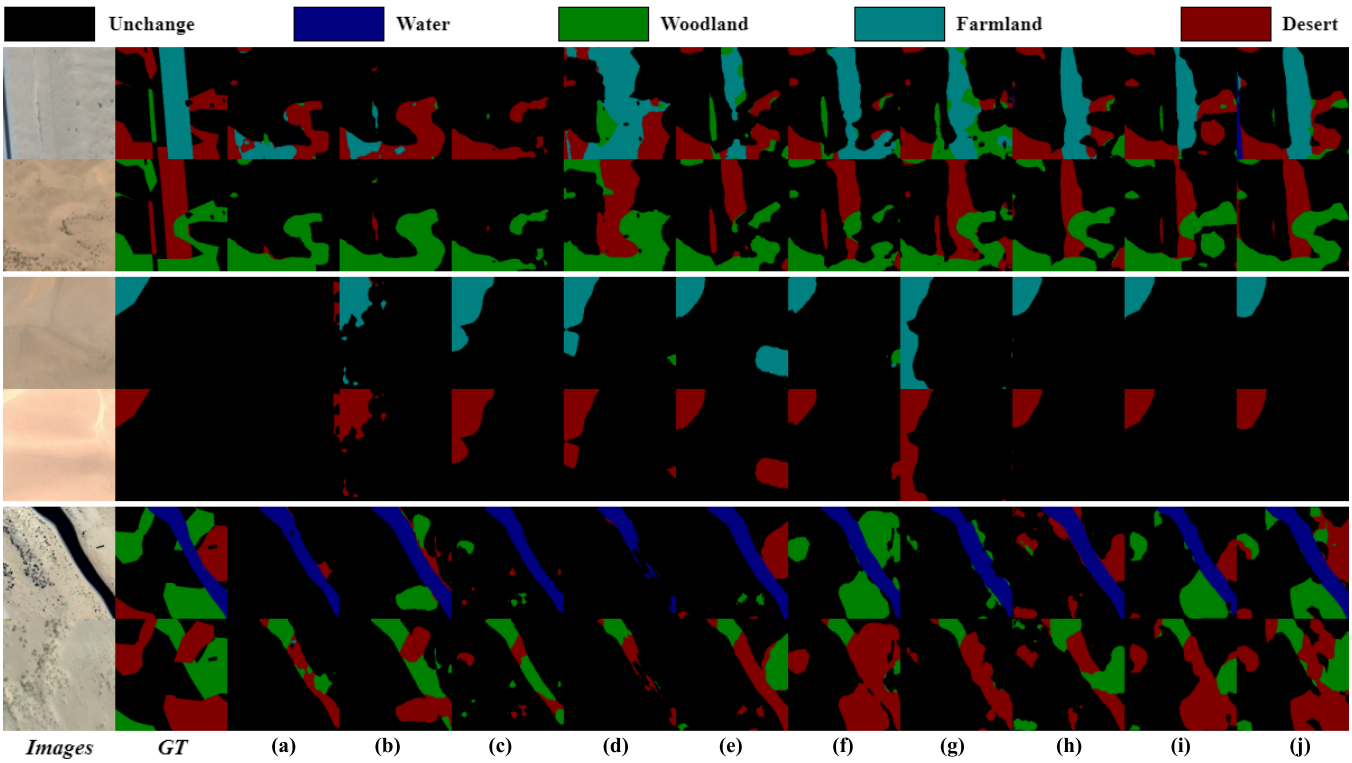


Fig. 6. Visualization of DSCD. (a)–(g) SSCDI [38], BiSRNet [38], SSESN [82], CdSC [39], MambaSCD [53], SCD-SAM [64], and SAM-FE [83], respectively. (h)–(j) Our proposed FILFSCD (ViT-H), FILFSCD (Tiny-ViT), and FILFSCD (Hiera-Large), respectively.

Among BCD models, USSFCNet exhibits the smallest parameter count. However, FILFBCD (Tiny-ViT) achieves the lowest Flops, VRAM, and inference latency across all evaluated models. This efficiency stems from the fact that

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TABLE V
ABLATION STUDY OF LORA'S RANK FOR BCD. THE BEST METRICS ARE MARKED IN BOLD

Rank	Params.(M)	VRAM(M)	MLCD				S2Looking				SYSU-CD			
			OA(%)	IoU(%)	Kappa(%)	F1(%)	OA(%)	IoU(%)	Kappa(%)	F1(%)	OA(%)	IoU(%)	Kappa(%)	F1(%)
r=0	**6.55**	**64.48**	90.27	66.16	73.25	79.63	99.23	47.98	64.52	64.84	91.19	67.37	74.83	80.51
r=4	6.62	64.75	91.63	70.08	76.93	82.41	99.17	49.14	65.27	65.89	91.68	69.98	76.90	82.34
r=8 (Ours)	6.67	64.97	91.78	71.74	78.06	83.34	**99.26**	**50.09**	**66.38**	**66.75**	92.13	71.69	78.34	83.51
r=16	6.76	65.43	91.82	71.73	78.38	83.51	99.24	49.75	66.06	66.44	92.33	71.60	78.47	83.45
r=64	7.51	68.15	**92.11**	**71.80**	**78.40**	**83.54**	99.24	49.84	66.37	66.53	**92.41**	**71.93**	**78.73**	**83.67**

TABLE VI
ABLATION STUDY OF LORA'S RANK FOR SCD. THE BEST METRICS ARE MARKED IN BOLD

Rank	Params.(M)	VRAM(M)	SECOND				DSCD			
			OA(%)	MIoU(%)	Sek(%)	F1(%)	OA(%)	MIoU(%)	Sek(%)	F1(%)
r=0	**6.71**	**73.54**	89.31	72.07	21.66	62.52	86.94	75.97	42.54	79.15
r=4	6.78	73.81	89.79	73.21	23.24	63.24	90.49	81.97	54.15	85.11
r=8 (Ours)	6.84	74.03	89.85	73.39	23.73	64.05	91.55	83.86	58.09	86.75
r=16	6.96	74.49	89.89	73.39	23.47	64.33	91.56	83.63	58.32	86.79
r=64	7.67	77.21	**90.01**	**73.41**	**23.97**	**64.37**	**91.66**	**83.94**	**58.57**	**87.02**

the weights of Mobile SAM are frozen during training and thus excluded from optimization. Consequently, despite its relatively large nominal parameter count, the effective computational load during both training and inference remains minimal. Both EFI-SAM and FILFBCD (Tiny-ViT) utilize Mobile SAM. Although EFI-SAM has slightly fewer Params. than FILFBCD (Tiny-ViT), its decoder employs an attention mechanism, resulting in higher Flops, VRAM, and Infer. latency compared to FILFBCD (Tiny-ViT). This demonstrates the superior efficiency of the FFT-based approach. As further corroborated by the results in Table I, FILFBCD (Tiny-ViT) consistently outperforms EFI-SAM across all metrics.

Among SCD models, although SCD-SAM and SAM-FE leverage Mobile SAM for feature extraction, their model efficiency is lower and its resource consumption is higher than that of FILFSCD (Tiny-ViT). FILFSCD (Tiny-ViT) demonstrates the most favorable efficiency profile among comparison SCD models, with the lowest VRAM of 0.07 GB, significantly lower than all competing approaches. By jointly analyzing the efficiency metrics in Tables III and IV alongside the performance results in Tables I and II, FILFArch effectively strikes an optimal balance between computational efficiency and performance, when applied to appropriately selected SAM series through parameter-efficient fine-tuning.

F. Ablation Studies of LoRA's Rank

In the FILFArch, we conduct an ablation study on the LoRA rank to investigate its impact on model performance and efficiency. Specifically, we compare models with varying ranks in terms of Params. (M), VRAM (M), and task-specific metrics. As shown in Tables III and IV, FILFArch (Tiny-ViT) exhibits relatively low parameter counts and minimal VRAM overhead, making it particularly suitable for discerning subtle differences introduced by different LoRA ranks.

Tables V and VI present the performance of models with different LoRA ranks on BCD and SCD tasks, respectively. The results reveal that model performance does not scale linearly with increasing rank. When the rank is increased from 0 to 8, $F1$ improves by 3.71%, 1.91%, 3%, 1.35%, and 7.60% on MLCD, S2Looking, SYSU-CD, SECOND, and DSCD, respectively, while the Params. increases by only 0.12 M on BCD tasks and 0.13 M on SCD tasks. However, further increasing the rank from 8 to 64, $F1$ improves merely 0.20%, 0.16%, 0.32%, and 0.27% on MLCD, SYSU-CD, SECOND, and DSCD, respectively. Meanwhile, this rank expansion incurs a substantial increase in model size, adding 0.84 M and 0.83 M in BCD and SCD tasks, respectively.

Consequently, although higher ranks generally lead to modest performance gains in most scenarios, the tradeoff between performance improvement and computational cost becomes unfavorable beyond rank = 8, and rank = 8 is used as the configuration for the proposed FILFArch.

G. Ablation Studies of Modules

The core components of the proposed framework include BIF-LoRA and a decoder composed of CDFE and FDFE. This section presents ablation studies conducted to validate each module's effectiveness. All experiments here are fine-tuned on the Mobile SAM.

Table VII presents the results of the ablation studies for the BCD task. "Baseline" denotes the configuration without fine-tuning SAM, employing a BCD Decoder that excludes CDFE and replaces FDFE with a standard ResBlock. "+CDFE" indicates only CDFE used in the BCD Decoder. "+FDFE" indicates only FDFE used in the BCD Decoder. "+C&F" indicates the use of both CDFE and FDFE in the BCD Decoder. "+LoRA" refers to fine-tuning SAM using conventional LoRA. "+BIFL" denotes fine-tuning SAM with the proposed BIF-LoRA. Fig. 7 illustrates the Grad-CAM visualization corresponding to Table VII.

Compared to the "Baseline," "+CDFE" achieves IoU improvements of 0.70%, 2.18%, and 2.86% on the MLCD, S2Looking, and SYSU-CD, respectively, while "+FDFE" yields gains of 0.39%, 1.34%, and 1.85% on the same

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TABLE VII
ABLATION STUDY FOR BCD. THE BEST METRICS ARE MARKED IN BOLD

Modules	MLCD				S2Looking				SYSU-CD			
	OA(%)	IoU(%)	Kappa(%)	F1(%)	OA(%)	IoU(%)	Kappa(%)	F1(%)	OA(%)	IoU(%)	Kappa(%)	F1(%)
Baseline	90.19	64.77	72.30	78.62	99.22	44.64	61.39	61.73	90.07	64.04	71.67	78.08
+CDFE	90.30	65.47	72.59	79.13	99.24	46.82	63.29	63.78	90.64	66.90	73.52	80.17
+FDFE	90.03	65.16	72.31	78.91	99.20	45.98	62.44	62.99	90.23	65.89	72.37	79.01
+C&F	90.27	66.16	73.25	79.63	99.23	47.98	64.52	64.84	91.19	67.37	74.83	80.51
+LoRA	91.02	67.38	74.71	80.51	99.26	46.49	63.37	63.47	91.31	67.92	75.28	80.90
+BIFL	91.35	69.07	76.07	81.71	99.22	48.03	64.55	64.89	91.36	68.43	75.64	81.25
+C&F+BIFL (Ours)	**91.78**	**71.74**	**78.06**	**83.34**	**99.26**	**50.09**	**66.38**	**66.75**	**92.13**	**71.69**	**78.34**	**83.51**

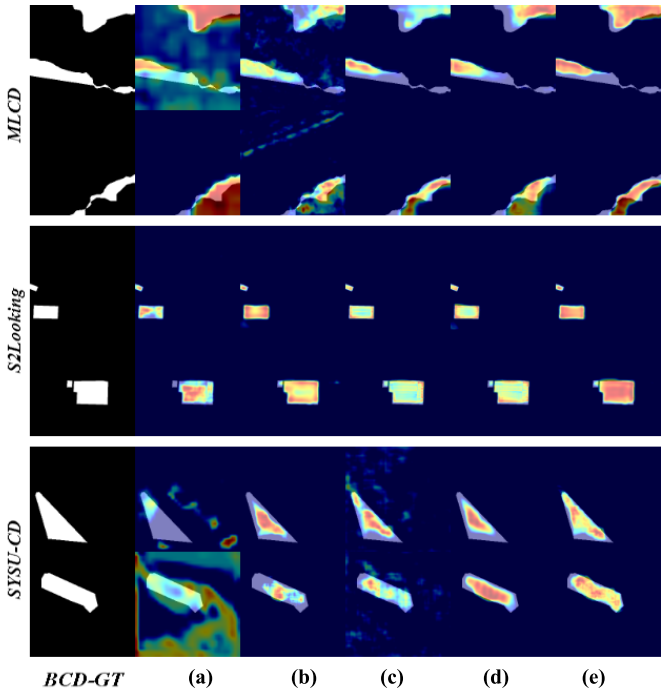


Fig. 7. Grad-CAM visualization of BCD ablation studies. (a)–(e) Baseline, +C&F, +LoRA, +BIFL, and FILFBCD, respectively.

benchmarks. These results demonstrate that either module alone can effectively enhance model performance. Moreover, "+C&F" configuration consistently outperforms both individual CDFE and FDFE modules, improving IoU by 1.39%, 3.34%, and 3.33% on the MLCD, S2Looking, and SYSU-CD, respectively, compared to the "Baseline." This confirms that frequency-domain guided fusion and enhancement of differential features effectively strengthen the aggregation of change-related information. Fine-tuning SAM with standard LoRA also yields improvements over the "Baseline," achieving a 2.61% IoU gain on MLCD. However, its performance on S2Looking falls slightly below that of "+C&F," indicating that parameter-efficient adaptation alone is not sufficient, and task-specific architectural enhancements are essential for robust BCD. The proposed BIF-LoRA delivers the most substantial performance gains across all datasets. On MLCD, IoU reaches 69.07%, surpassing the "Baseline" by 4.30% and outperforming standard LoRA by 1.69%. Similar trends are observed on SYSU-CD and S2Looking.

Through Grad-CAM visualization, it can be intuitively observed that BIF-LoRA more effectively attends to regions with changes and localizes the target object with greater accuracy. This highlights that the FDFI significantly enhances the ability of SAM to learn consistent cross-temporal representations in RSIs, thus better aligning its general-purpose vision priors with the demands of the BCD task. In particular, FILFBCD achieves the best results across all metrics and datasets, underscoring the synergistic effect between FDFI and the enhanced decoding pathway.

To further validate the contribution of each module within the SCD task, we conducted repeated ablation experiments on both the SECOND and DSCD. The results are presented in Table VIII, where the symbol definitions remain consistent with those in Table VII. In particular, within the SCD tasks, FDFE is also utilized for semantic feature recovery.

On the SECOND, "+CDFE" improves MIoU by 0.29% and Sek by 0.79%, while "+FDFE" improves MIoU by 0.64% and Sek by 0.74%. When both CDFE and FDFE are combined ("+C&F"), the model achieves an MIoU of 72.09% and Sek of 21.66%. Overall, FDFE demonstrates a greater individual improvement over CDFE, and their combination further enhances model performance. The same trend is also observed on the DSCD. This indicates that, in SCD tasks, leveraging frequency-domain features facilitates more reliable detection of subtle semantic changes. Meanwhile, on the DSCD, the effectiveness of BIF-LoRA for the SCD tasks is particularly evident as it achieves MIoU improvement of 6.81% and Sek increase of 12.53% compared to the "Baseline." Compared to standard LoRA, BIF-LoRA shows the gains of 3.6% in MIoU and 6.31% in Sek. Fig. 8 presents Grad-CAM visualizations corresponding to BCD results, providing intuitive evidence that CDFE, FDFE, and BIF-LoRA collectively enhance cross-temporal discrepancy feature reasoning capabilities.

Fig. 9 presents the "from–to" transition matrix from the ablation studies and provides the detailed information about semantic changes. By comparing (b)–(d), it is evident that BIF-LoRA, leveraging its dual-temporal feature interaction mechanism, extracts land cover semantic information more effectively than the original SAM. Concurrently, the integration of FDFE with BIF-LoRA substantially enhances the recovery capability of semantic features. The "from–to" transition matrix clearly demonstrates that FILFSCD achieves superior performance in SCD tasks by accurately modeling both the spatial coherence and temporal evolution dynamics of land cover changes.

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TABLE VIII
ABLATION STUDY FOR SCD. THE BEST METRICS ARE MARKED IN BOLD

Modules	SECOND				DSCD			
	OA(%)	MIoU(%)	Sek(%)	F1(%)	OA(%)	MIoU(%)	Sek(%)	F1(%)
Baseline	88.71	71.04	19.86	60.29	85.80	73.53	37.97	76.34
+CDFE	98.44	71.33	20.65	60.92	85.89	74.06	39.03	77.04
+FDFE	89.06	71.68	20.63	61.44	86.47	75.18	40.30	77.67
+C&F	89.31	72.07	21.66	62.52	86.94	75.97	42.54	79.15
+LoRA	88.98	72.03	21.73	62.05	87.44	76.74	44.19	80.22
+BIFL	89.15	72.26	22.06	62.45	89.47	80.34	50.50	82.91
+C&F+BIFL (Ours)	**89.85**	**73.39**	**23.73**	**64.05**	**91.55**	**83.86**	**58.09**	**86.75**

TABLE IX
COMPARISON OF FEATURE DOMAIN FOR BCD. THE BEST METRICS ARE MARKED IN BOLD

Methods	Params.(M)	Flops(G)	VRAM(M)	MLCD				S2Looking				SYSU-CD			
				OA(%)	IoU(%)	Kappa(%)	F1(%)	OA(%)	IoU(%)	Kappa(%)	F1(%)	OA(%)	IoU(%)	Kappa(%)	F1(%)
ResBlock	11.12	5.04	71.58	90.91	67.64	74.77	80.70	99.24	47.07	63.64	64.01	91.08	68.38	75.37	81.22
Transformer	45.30	30.56	238.12	91.35	68.70	75.85	81.45	99.22	48.79	65.02	65.61	92.00	70.29	77.38	82.56
Mamba	28.20	18.23	236.37	90.93	67.73	74.85	80.76	99.20	48.53	64.76	65.35	91.77	70.09	77.04	82.41
FFTB(Ours)	**6.67**	**3.64**	**64.97**	**91.78**	**71.74**	**78.06**	**83.34**	**99.26**	**50.09**	**66.38**	**66.75**	**92.13**	**71.69**	**78.34**	**83.51**

TABLE X
COMPARISON OF FEATURE DOMAIN FOR SCD. THE BEST METRICS ARE MARKED IN BOLD

Methods	Params.(M)	Flops(G)	VRAM(M)	SECOND				DSCD			
				OA(%)	MIoU(%)	Sek(%)	F1(%)	OA(%)	MIoU(%)	Sek(%)	F1(%)
ResBlock	11.94	6.02	93.48	89.34	72.58	22.38	62.68	90.05	81.08	51.96	83.94
Transformer	46.27	31.62	243.40	89.73	72.94	22.97	63.28	90.77	82.43	54.88	85.33
Mamba	28.68	19.03	241.47	89.44	72.92	23.28	63.66	90.49	82.05	54.24	84.93
FFTB(Ours)	**6.84**	**4.76**	**74.03**	**89.85**	**73.39**	**23.73**	**64.05**	**91.55**	**83.86**	**58.09**	**86.75**

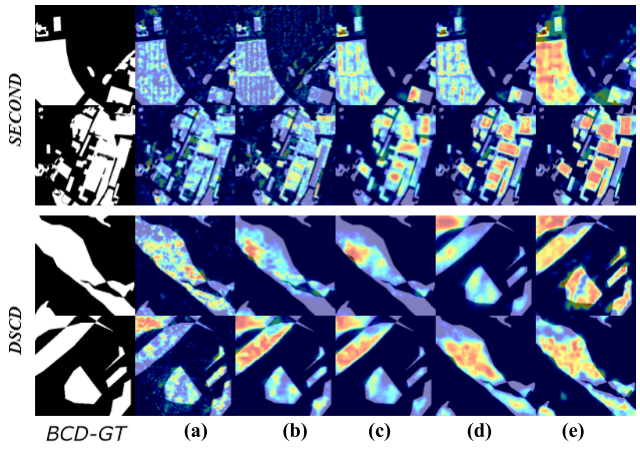


Fig. 8. Grad-CAM visualization of SCD ablation studies. (a)–(e) Baseline, +C&F, +LoRA, +BIFL, and FILFSCD, respectively.

V. DISCUSSION

A. Feature Domain and Model Performance

To validate the effectiveness of frequency-domain information in RSCD and to substantiate the necessity of the proposed frequency-domain interactive fine-tuning algorithm, we conduct comparative experiments using four distinct feature modeling paradigms: ResBlock, Transformer, Mamba, and FFTB (Ours). These correspond to modeling spatial local features, spatial-global relationships, sequential-temporal dynamics, and frequency-domain representations. This design enables a comprehensive comparison among spatial, sequential, and spectral representations in RSCD tasks. To ensure a thorough evaluation, we not only compare the task performance of these methods but also report their computational efficiency, including Params., Flops, and VRAM. All metrics are computed under an input resolution of 256×256. Tables IX and X present the experimental results on BCD and SCD tasks, respectively.

As shown in Table IX, frequency-domain features demonstrate clear advantages in BCD tasks because they are highly sensitive to complex textural changes, low-frequency structural discrepancies, and cross-view geometric variations, phenomena that are often challenging for purely spatial models to capture. FFTB achieves the highest performance across all three BCD datasets. It does so with the lowest Params., Flops, and VRAM among all compared methods. Although Transformer-based yields competitive accuracy, its VRAM is nearly four times higher than that of FFTB, highlighting the efficiency gap. Similarly, in Table X, the FFTB approach consistently achieves the best performance while maintaining significantly lower computational costs; the Params., Flops, and VRAM of FFTB are substantially reduced compared to both Transformer and Mamba. This confirms that frequency-domain representations are not only effective for detecting structural changes but are also highly discriminative for SCD tasks.

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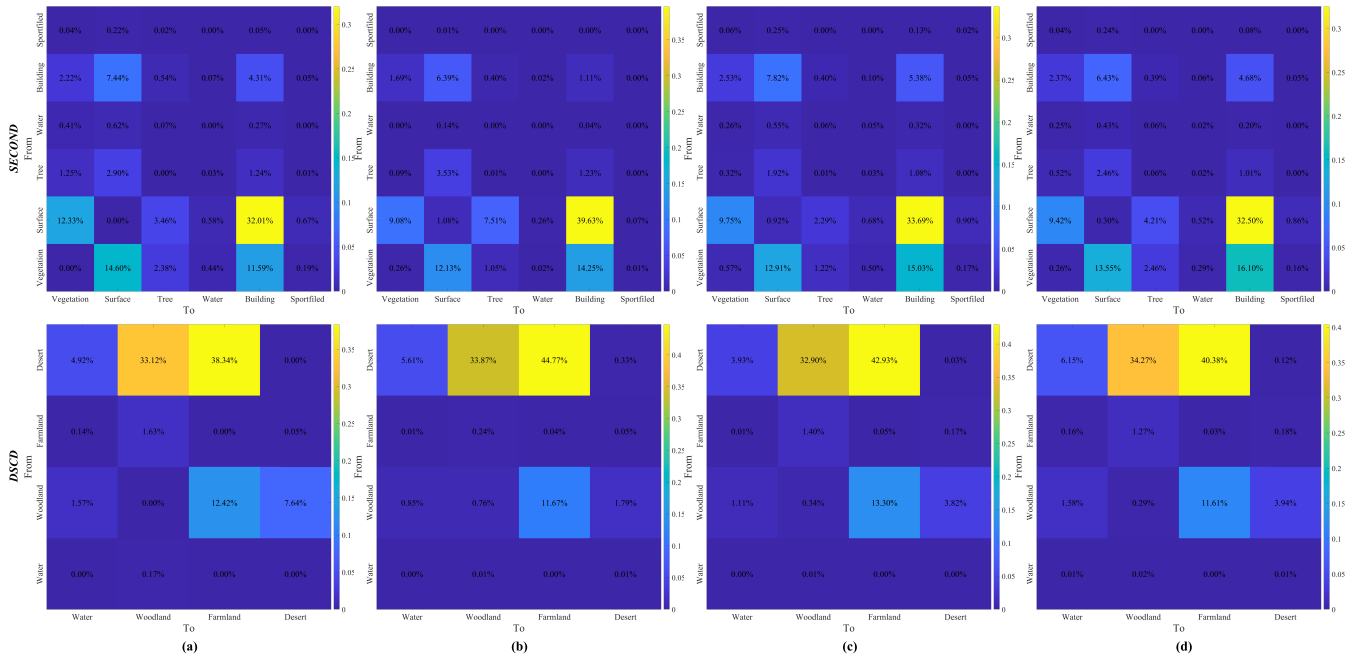


Fig. 9. Semantic change matrix providing "from–to" transition information derived from the ablation experiment results. (a)–(d) Reference, Baseline, +BIFL, and FILFSCD, respectively.

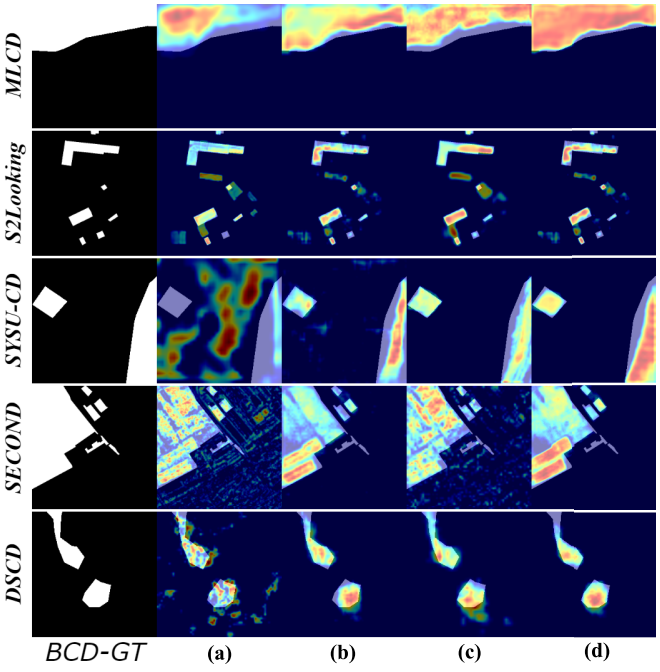


Fig. 10. Grad-CAM of difference feature domain comparison. (a)–(d) ResBlock, Transformer, Mamba, and FFT (Ours), respectively.

Collectively, these results strongly support the efficacy and efficiency of incorporating frequency-domain information into SAM fine-tuning for RSCD, and they underscore the advantage of our proposed frequency-domain interaction mechanism over conventional spatial or sequential modeling strategies.

To intuitively compare the effectiveness of different modeling strategies in RSCD, Fig. 10 visualizes the Grad-CAM maps produced by each method on BCD tasks. As illustrated in Fig. 10, the FFT method demonstrates a superior localization capability, consistently highlighting regions that genuinely undergo change more precisely than competing methods.

Specifically, on the S2Looking, FFT exhibits notably stronger and sharper activations along the boundaries of small-scale changed objects, outperforming other approaches in capturing fine-grained structural details. On the DSCD, ResBlock produces diffuse and scattered Grad-CAM responses, indicating insufficient focus on true change regions. While Transformer and Mamba localize changes with more attention than ResBlock, their activations along object boundaries remain less pronounced compared to FFT. These Grad-CAM further corroborate that frequency-domain representations enhance the model's sensitivity to subtle changes, particularly for small targets and edge structures. Grad-CAM analysis thus provides compelling qualitative evidence that integrating frequency-domain information enables more accurate and focused change localization in RSCD tasks.

Fig. 10 has already demonstrated the advantage of frequency-domain features over spatial- and temporal-domain features in detecting change regions. To further illustrate the relationship between frequency-domain features and change information, Fig. 11 presents the FFT-based frequency-domain distribution of change features along with the low- and high-frequency features of the prediction results of FIFLArch.

As shown in Fig. 11, the BCD-GT spectrum exhibits strong consistency with the predicted spectrum. In scenes involving prominent structural changes, such as buildings and roads, the spectral energy shows a clear concentration within specific frequency bands. Furthermore, the physical meaning of the weighted frequency-domain features is clearly demonstrated in the decomposition of change information. Low-frequency components physically correspond to the continuous semantic

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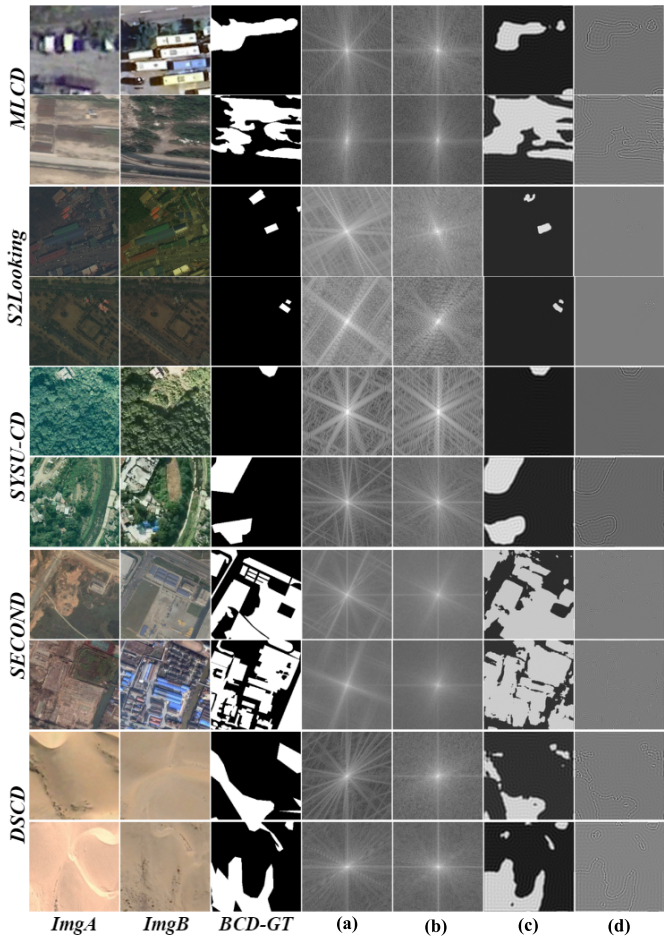


Fig. 11. Visualization of frequency-domain features and change information. (a)–(d) FFT magnitudes of BCD-GT, FFT magnitudes of BCD-GT of prediction, low-frequency features of prediction, and high-frequency features of prediction, respectively.

structures and global illumination of the scene. As shown in Fig. 11, regions undergoing extensive construction or land cover transitions exhibit pronounced low-frequency responses, indicating that the learnable weights successfully prioritize these components to capture overall scene geometry and broad-area alterations. In contrast, high-frequency components physically represent rapid signal variations, such as fine-grained boundaries and texture patterns. The model's weighted interaction acts as an adaptive filter here, amplifying these specific frequencies to recover sharp edges in Fig. 11(d) while suppressing the high-frequency noise typically inherent in complex remote sensing backgrounds. Subtle structural changes, such as boundaries, road edges, and small structures, generate distinct responses in the high-frequency range. These high-frequency features align closely with fine boundaries in the change maps, further demonstrating the model's capability to localize spatially detailed changes. In essence, RSCD encompasses both low-frequency features and high-frequency features. Fig. 11 further illustrates that frequency-domain analysis can effectively disentangle these distinct change features.

B. Limitations and Improvements

Despite the notable performance improvements achieved by FILFArch both on the BCD tasks and SCD tasks, several limitations remain that require focused attention in subsequent work.

First, FILFArch exhibits insufficient sensitivity to subtle targets under complex backgrounds. Although it demonstrates superior localization and classification capabilities for changed targets compared to other methods, it struggles with gradual changes occurring within intricate scenes, a limitation particularly evident in SCD tasks. For instance, in the SECOND, vegetation and trees are easily confused. Similarly, in the DSCD dataset, the desert often masks farmland. Such changes typically induce only minor shifts in the space features, which can be readily overwhelmed by noise or inconsistencies in the RSIs. This issue is pronounced when appearance differences are minimal, yet semantic distinctions are significant, as frequency-domain features may fail to effectively discriminate between visually similar classes. Second, in urban land cover imagery, most objects exhibit highly similar edge and texture features. Texture surfaces generate strong high-frequency signals that are easily misinterpreted as meaningful changes, leading to false classification. This challenge is prominently observed in both the MLCD and SECOND. Third, FILFArch relies on the feature extraction capability of SAM. As Transformer-based VFMs, SAM prioritizes global spatial contextual correlations during feature extraction, whereas FILFArch employs FDFIs. The misalignment between spatial-domain representations and frequency-domain information constrains the model's ability to capture consistent cross-temporal features, thus affecting its detection of subtle changes. Fourth, current evaluations of FILFArch are exclusively based on optical imagery, leaving its generalization performance under heterogeneous sensor modalities and extreme imaging conditions unverified.

To address these challenges, future research will focus on the following directions. First, we aim to develop an adaptive frequency-domain selection mechanism that integrates multiscale spatial-frequency features. This will enhance the model's sensitivity to spectral ranges critical for detecting subtle semantic changes. Second, contrastive learning can be leveraged to disentangle edge and texture representations from semantic content, thereby learning more robust and discriminative embeddings. Third, a unified framework that enables mutual guidance between spatial-domain feature extraction and frequency-domain transformation can be proposed to resolve feature domain misalignment. Finally, future work will extend validation to heterogeneous data sources by developing cross-modal frequency-domain adaptation mechanisms, ensuring robustness across diverse sensing modalities.

VI. Conclusion

This article proposes FILFArch, a novel frequency-domain interactive LoRA architecture to fine-tune SAM in RSCD tasks. Building upon FILFArch, we design FILFBCD and FILFSCD for BCD and SCD tasks, respectively. Both FILFBCD and FILFSCD utilize SAM to extract features from bi-temporal RSIs. The BiF-LoRA is designed to fine-tune the SAM, enhancing its capability to capture cross-temporal feature relationships. Within BiF-LoRA, the FDFI employs

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FFT to facilitate interaction between bi-temporal frequency-domain features, aiming to extract temporal differences. The BCD Decoder, as a binary change decoder, serves both FILFBCD and FILFSCD. The BCD Decoder first utilizes CDFE to perform an initial fusion of bi-temporal deep semantic features, generating a coarse-grained change map. Subsequently, the FDFE refines these abstract features to produce a fine-grained change map. In FILFSCD, FDFE is also employed for recovering semantic change information from the bi-temporal RSIs. To explore the potential of FILFArch in RSCD tasks, experiments were carried out using SAM, Mobile SAM, and SAM2 as the encoder. All variants of the FILFArch achieve superior performance in RSCD tasks.

Experimental evaluations on multiple BCD and SCD datasets demonstrate that FILFArch achieves superior performance over existing RSCD methods, confirming its effectiveness in capturing fine-grained changes and semantic transitions in land cover. Ablation studies further validate the contributions of each module within the FILFArch, highlighting the importance of FDFI. Overall, FILFArch provides a robust and efficient solution for adapting SAM to the unique challenges of bi-temporal RSCD tasks.

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